

“Many of the players in this industry are primarily importers and distributors of educational and hobby products. The products are imported without paying customs duty and other taxes and sold without proper paper work. There is no quality assurance. These bad practices will be the biggest killers of the Indian market.” —Sachitanand Malewar, director, Nex Robotics

less communication, power electronics, instrumentation, VLSI, VHDL and digital communications.”

Deshpande feels the need for a different range of products to cater to different demographic sections. In fact, according to him, schools have the potential to become a bigger end user of educational kits than engineering colleges. He also emphasises on the need to explore the market associated with Tier-II and Tier-III engineering colleges.

Competitive advantages and challenges

The increased investments in educational sector have attracted electronic design SMEs to the educational kits industry. With an increase in the demand for educational kits, they have started exporting them as well while serving the local market. In due course, the big players too have understood the value

of this market and increased their level of penetration in India.

Venkat lists out some competitive advantages of India as availability of multiple high-end real-time kits, capabilities in mixed-signal test (which makes India an ideal place for developing full system solutions), cost competitiveness and fast-growing local market.

“The skilled labour required for this industry is easily available in India. Also, in India prices are very competitive compared to many other developed countries,” adds B.M. Shah, proprietor, Edkits Solution. India is strategically located, which provides the opportunity for export to the neighbouring countries.

Coming to challenges, Deshpande says, “The public opinion, although declining, is that robots are machines that replace humans. That attitude can be very dangerous to a nation like ours. I feel that is one of the important reasons why robotics hasn’t made it big in education.” He also points out that “Expertise is in high demand globally. Coming up with incentives to retain the existing staff and developing new staff is a problem that needs immediate attention.” The hobby kits industry is, in general, price-sensitive, which is another challenge to overcome.

According to Joshi, tax burden, be it excise or high rate of VAT, is a major bottleneck for the growth of the industry. Tax relief to the industry would help in effective cost-control and therefore large-scale adoption of newer technologies.

Venkat says, “In this competitive scenario where umpteen number of companies are coming up with their respective kits, it is difficult for the col-

“Presently, the ecosystem needs to focus on better coordination, holistic approach to providing solutions to the educational requirements and introduction of affordable products so that students can have their own personal labs.” —Thushar V. Nadh, director, rhydoLABZ

leges, students and hobbyists to decide as to which resource kit is actually more effective and result-driven.”

Need to move up the value chain

Currently, the industry is in the middle of the value chain. Most companies do little in-house development and need to import the items. It’s high time that the industry looks for options and ventures into high-return verticals too.

“There are various ways in which Indian companies can move up the value chain. I feel consulting and ICT is the way out. As we have substantial domain knowledge of ICT, we can provide solutions to the world,” feels Joshi.

Joshi adds, “Time has come to think of solutions first and provide them to the world in the form of ready products. Pros of product development, although a high-risk business, far outweigh the cons. By not getting into

“Lack of vision is a common problem amongst educational and hobby kits players. They should improve the manufacturing process by following the global standards. At the same time, they should also focus on after-sales support and service.” —Rajesh Sawant, director, Falcon Electrotech

